

# Defining Stakeholder Relationships

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**J**ointly developed by the ISO and IEC in 1995, the ISO/IEC 12207 standard, *Software Life Cycle Processes*, provides specific guidance in defining the roles and responsibilities of various stakeholders in the life cycle of a software project, product, or service. And the software community is beginning to take heed. James W. Moore, a leading expert on software engineering standards, has indicated the importance of this standard (*Software Engineering Standards: A User's Road Map*, IEEE CS Press, Los Alamitos, Calif., 1997):

ISO/IEC 12207 may be the most important software engineering standard ever written. Within the Information Technology industry, the standard may have impact similar to that of the "ISO 9000" series of standards. Already, the standard is being adopted by influential organizations within the industry.

The standard itself is relatively brief, detailing 17 processes in less than 40 pages. The 17 processes are divided into three main process groups:

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Defining stakeholder roles and responsibilities in the life cycle of a project, product, or service is crucial.

- *Primary processes* include acquisition, supply, development, operation, and maintenance.
- *Supporting processes* include documentation, configuration management, quality assurance, verification, validation, joint review, audit, and problem resolution.
- *Organizational processes* include management, infrastructure, improvement, and training.

Each process breaks down into relevant activities and tasks that reflect a clear plan-do-check-act cycle. One further process, tailoring, specifies the activities and tasks to follow in adapting the standard to a particular situation or application.

But just how useful is this standard?

Based upon personal mentoring experiences, I briefly describe two case studies that demonstrate the standard's importance and versatility:

- Haldex Traction AB of Landskrona, Sweden, the supplier of a safety-critical automotive component to Volkswagen GmbH, among others, and
- Cambio+ Healthcare Systems AB of Linköping, Sweden, the supplier of a medical information system for hospitals and clinics.

In these two cases, ISO/IEC 12207 has provided important value-added guidance to the stakeholders involved in each project. Drawn from very different industries, these case studies also show that ISO/IEC 12207 can be applied to both systems and software engineering.

## AUTOMOTIVE SYSTEMS ENGINEERING

Haldex Traction AB has developed a new coupling device for four-wheel vehicles: the Limited Slip Coupling (LSC). The LSC dynamically adapts the degree of stiffness in the rear-axle coupling based on several parameters including speed, temperature, braking, turning, and tire and road conditions. This small electromechanical device resides in a harsh environment and is safety-critical. LSC represented an important new market to Haldex. The first series installations of LSC products will be in Volkswagen automobiles.

The LSC project involved a multidisciplinary effort among mechanical, electronic, systems, application, and software specialties. Haldex AB involved several subcontracting stakeholders in the process. The product evolved over time as a series of prototypes that engineers used in simulated and live testing in various parts of the world.

The project leader realized the importance of structuring the project around a sound set of processes that would guide Haldex and the other stakeholders involved. Given that no satisfactory system life cycle process standard existed at that time, the project leader, based upon my recommendation, decided to use ISO/IEC

12207 for the system software and application software. We then decided to define and *tailor* additional processes to cover the mechanical and electronic aspects. Thus, in this application, Haldex AB deployed ISO/IEC 12207 as a system standard.

The project group used the standard in the initial stages to establish the project structure, in particular for identifying stakeholders and their capabilities, roles, and responsibilities. Since the product is safety-critical, it was vital to establish a documentation structure that would capture the essential aspects of the product as well as the project. This document structure is shown in Figure 1.

As noted in the figure, the top-level document is a safety case. This document summarizes the information needed to provide certain evidence about qualified stakeholders. It shows that stakeholders developed the product with suitable product technologies and according to defined processes and a safety plan.

The process document defines tailored versions of the ISO/IEC 12207 primary processes. Since most of the supporting processes are vital from a safety point of view, tailored versions of these processes were incorporated into the safety plan.

Haldex AB further tailored the standard to meet the needs of the product development model, a model that I can best describe as “incremental builds with concurrencies”—incremental within each prototype and concurrent in that multiple prototype and series versions were developed and tested simultaneously.

The importance and usefulness of ISO/IEC 12207 was several-fold. While project pressures caused deviations from the established processes, the set of base documents provided vital assistance in establishing and communicating the project framework. Those base documents also helped in orienting new personnel, as well as provided an important sales argument for Haldex AB, which now has sold the LSC product to another major automotive manufacturer.

### HEALTH CARE SOFTWARE

Cambio+ Healthcare Systems AB, a small but rapidly developing company, develops medical information processing

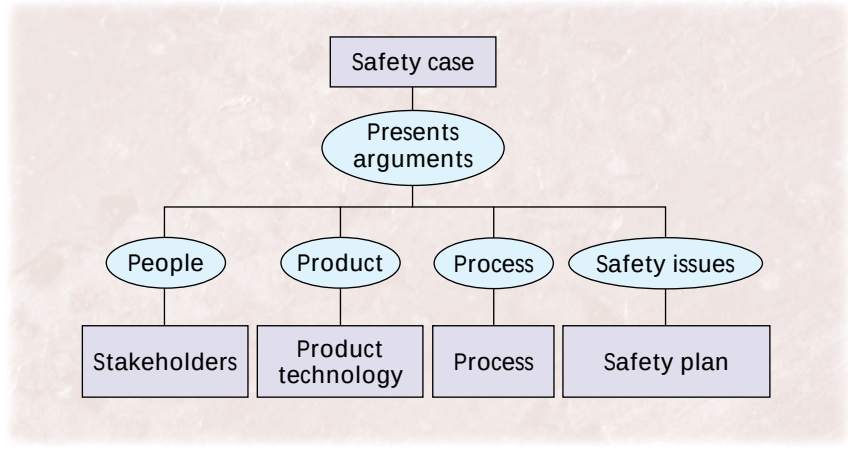


Figure 1. Document structure for the LSC product.

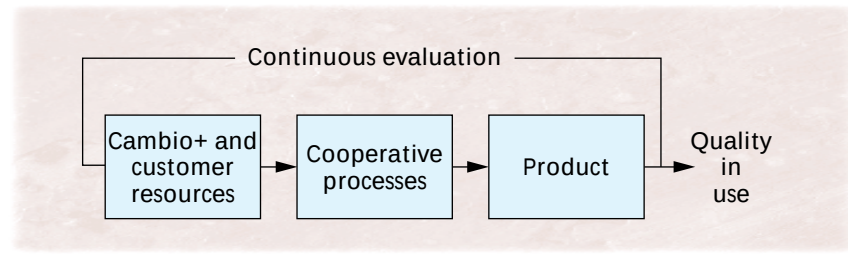


Figure 2. A continuous supply and feedback loop for Cambio+ software development.

products for hospitals and clinics. The functions supported include both inpatient and outpatient service, administration and patient billing, scheduling of medical services (including operations), laboratory results, pharmacology, and medical specialty journals.

In Sweden and elsewhere, significant problems have arisen in deploying medical information processing systems in complex hospital environments. Many suppliers attempt to provide major systems in a “big bang.” That is, they provide large, complex, often inflexible systems at the initial installation.

Unfortunately, this has resulted in costly problems for the customers, often to the point of triggering legal actions. Problems involve missed requirements, user dissatisfaction, inadequate performance and reliability, inadequate training, and the inability of customers to influence the system’s development.

Recognizing these problems, Cambio+ concentrated on developing the right tech-

nology as well as the right customer-related processes to ensure successful installation and support of their products in clinics as well as complex hospital environments. To this end, Cambio+ developed a business strategy based on the application of ISO/IEC 12207 as well as ISO/IEC 9126 (product quality characteristics). The basic premise of the strategy is to provide “Quality in Use” by achieving the following product objectives:

- meet the customer’s needs,
- be reliable, and
- be developed in a manner that permits active customer participation in product evaluation and improvement.

To achieve these quality goals, Cambio+ views each customer relationship as a cooperative project between Cambio+ and its customer. Further, to avoid “big bang” problems, a progressive acquisition approach to purchasing

Cambio+ products and services is defined. This results in a continuous supply and feedback loop, as shown in Figure 2.

As indicated in the figure, Cambio+ and the customer supply resources (including personnel) and deploy them according to a set of ISO/IEC 12207-defined cooperative processes. This creates a product version, which is then verified and validated according to a set of product quality attributes following ISO/IEC 9126. After product deployment, and based on experience with the product, the customer and Cambio+ jointly evaluate the product and establish new goals for improving the existing version as well as adding new functionality. This carries them along in the development cycle, revisiting the resource aspect, the processes, and the product quality.

The cooperative processes are based on a first-level tailoring of relevant ISO/IEC 12207 processes down to the activity level, which concretely identifies the project roles and responsibilities of Cambio+ and the customer. This tailor-

ing serves as a firm basis for negotiation between Cambio+ and their customers. Further tailoring of the activities and task definition then provides a firm basis for project scheduling. Cambio+ can then employ the six major product quality attributes of ISO/IEC 9126 in verifying and validating the product: functionality, reliability, usability, efficiency, maintainability, and portability.

Cambio+ has, by using the ISO/IEC 12207 and 9126 standards, developed the backbone of a viable business concept that is gaining wide acceptance among its customers and potential customers.

**T**hese examples demonstrate that learning to use ISO/IEC 12207 constructively provides important value-added returns on investment. Already translated into French, Japanese, and Portuguese, the standard will soon be available in Swedish. A US adaptation, the standard has now been officially approved as IEEE/EIA 12207. In addition to containing the ISO/IEC standard, further guidance is provided in respect to documentation and utilization. The 12207 standard will continue to evolve, and the standard working group is scheduled to amend it in 2000 and then revise a few years after. In addition, the same working group is developing ISO/IEC 15288 (System Life Cycle Processes), which provides similar guidance for the life cycles of complex systems involving hardware, software, and humans. Given the fact that quality management systems based on processes occupy the current line of thinking for ISO 9001:2000, both ISO/IEC 12207 and eventually 15288 will play an increasingly important role in defining processes to be evaluated and certified. ♦

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### Resources

Copies of the ISO/IEC 12207 standard are available from national standards bodies, such as ANSI (Customer Services, 11 West 42nd St., New York, NY 10036), from ISO (1, rue de Varembe, CH-1211 Geneva 20, Switzerland), or IEC (3, rue de Varembe, CH-1211 Geneva 20, Switzerland).

The IEEE/EIA Std 12207 Software Life Cycle Processes adoption of the international standards is available from IEEE Standards Online <http://standards.ieee.org/catalog/olis/index.html>.

The following book thoroughly presents software engineering standards, including 12207:

S. Magee and L.L. Tripp, *Guide to Software Engineering Standards and Specifications*, Artech House, Boston, Mass., 1997.